

RAVITEJA MADDELA

MSc Computer Science & Engineering Student

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PROFESSIONAL SUMMARY

MSc Computer Science student at the Technical University of Denmark (DTU) with software development experience across healthcare, insurance, and financial domains (GE Healthcare, Swiss Re, American Express). Focused on machine learning, web applications, and software engineering.

EDUCATION

Technical University of Denmark (DTU) 2025 – Present
Master of Science in Computer Science & Engineering (Copenhagen, Denmark)
Key Focus: Machine Learning Systems, Multi-Agent Systems, Algorithm Engineering, Distributed Systems. Student ID: s252744.

RVR & JC College of Engineering (Acharya Nagarjuna University)
Bachelor of Technology in Information Technology (India)

TECHNICAL SKILLS

AI & ML Systems: PyTorch, QLoRA, Transformers, RAG, Mixture-of-Experts (MoE), BioClinicalBERT, FAISS

Backend & Web: Python, Java, JavaScript, Leaflet.js, Mapbox, FastAPI, Spring Boot, REST APIs, React, Tailwind CSS

Cloud & Tools: Docker, Git, Linux, AWS, Kaggle, HuggingFace, Netlify

FEATURED ENGINEERING & RESEARCH PROJECTS

Sparse MoE-QLoRA for Danish Legal QA PyTorch MoE QLoRA Personal Research Project

- Built a 4-bit Quantized Low-Rank Adaptation (QLoRA) pipeline to fine-tune a 7B model for Danish legal question answering.
- Implemented centroid-based Sparse Mixture-of-Experts (MoE) routing using E5 embeddings paired with RAG document retrieval.
- Tested memory pre-caching and in-place tensor copy to enable fast expert adapter switching on a single T4 GPU.

Weather-Aware Route Planner & Travel Advisor JavaScript Leaflet.js Mapbox Personal Project

- Built an interactive route planning web application combining Leaflet.js and Mapbox GL JS map rendering.
- Integrated OpenWeatherMap APIs to sample weather conditions along route waypoints in real time.
- Implemented Netlify proxy functions for secure API key handling, with planned integration of LLM API services for route weather safety advice.

Multi-Agent Coordination in Hospital Domain Java CBS A* Search DTU 02285 Course Project

- Developed a Java multi-agent path finding solver for autonomous robot agents in the MAVIS Hospital Domain grid world.
- Implemented Hungarian matching for box-goal allocation, fast greedy priority orderings, and 3-phase box staging.
- Built a Conflict-Based Search (CBS) repair engine with CAT bias, opportunistic bypass, Lemma-1 jump points, and segment replanning.

Multi-Modal Clinical Phenotyping System PyTorch FastAPI FHIR R4 Healthcare AI Project

- Built a dual-encoder PyTorch pipeline pairing BioClinicalBERT for unstructured clinical notes with a transformer for lab test features on MIMIC-IV patient records.
- Developed a FastAPI inference service with SHAP lab feature importance, BERT attention token scores, FHIR R4 bundle adapters, and Gradio web interface.

PROFESSIONAL EXPERIENCE

Software Developer
GE Healthcare (Bengaluru, India)

- Developed internal license management software for hospital equipment (MRI and CT scanners), enabling staff to activate and update feature licenses.
- Built automated installation and upgrade tooling to reduce manual overhead for active hospital scanners.
- Collaborated with engineering teams to deliver annual software feature updates and security patches.

Software Developer

Swiss Re (Hyderabad, India / Zurich, Switzerland)

- Developed *SwiftMX*, a transaction conversion tool mapping business team requests into automated bond and security purchase orders via third-party financial tools.
- Built testing and data migration utilities for TRAX application upgrades to verify configuration transfer success rates.
- Implemented custom encryption middleware routing payment files (salary and insurance disbursements) securely to banking institutions.

Junior Software Engineer

American Express (Hyderabad, India)

- Developed online PDF credit statement generation features for American Express Mexico internet banking.
- Participated in software development lifecycle: requirements review, component development, testing, and production deployment.

CERTIFICATIONS & LANGUAGES

Certifications: Python Institute Certified Programmer | Lean Six Sigma Yellow Belt | TOEFL iBT (Score: 94/120)

Languages: English (Fluent), Telugu (Native), Danish (Prøve i Dansk 2 / PD2)